

1

1.1 Background to the Problem

This platform helps clients quickly find skilled service providers nearby. Clients can create accounts and request services like plumbing, electrical work, or repairs by entering their location and needs. The system then matches them with available, qualified workers in the area. Workers can register, list the services they offer, and set the areas they cover so local clients can find them. Admins can manage user accounts, approve new workers, update services, and handle disputes using a dashboard. Together, these features create a safe, easy-to-use system for fast local service matching

1.2 Solution to the Problem and Process Model Selection

Project Scope

- A simple platform where anyone can book home services easily.
- Every service provider will be properly checked and verified for trust.
- Secure payment options so customers and providers feel safe.
- Mobile apps for both customers and service providers to use on the go

Feature list

◆ Client Features

1. **Client Registration/Login**
2. **Profile Management**
3. **Service Request**
4. **Worker Matching**
5. **Match Notification**
6. **Booking Request**

7. Helpline

◆ Worker Features

1. **Worker Registration/Login**
2. **Service Profile Setup**
3. **Availability Management**
4. **View Service Requests**

◆ Admin Features

1. **Admin Login**
2. **User Management**
3. **Service Management**
4. **Registration Approval**
5. **worker Monitoring**

◆ Clerk Features

1. **Clerk Login**
2. **Support Ticket Handling**
3. **Worker Monitoring**
4. **Worker Profile Verification**
5. **Communication Liaison**
6. **Data Entry & Updates**
7. **Report Drafting**
8. **Feedback Collection**

User Story Table

As A/An	I want to	So that	Acceptance Criteria
Customer	Sign up and log in securely	I can use the platform	Can register and log in with email/password
	Manage my profile	My info stays current	Can view and edit profile details
	Enter my service needs and location	I can find the right help	System accepts and validates service/location inputs
	See matched workers and get updates	I know who's available	System shows worker matches and sends notifications
	Book a specific worker	I can hire someone I like	Can send booking requests to chosen workers
	Get help or report issues	Problems get solved	Can submit support requests or complaints

As A/An	I want to	So that	Acceptance Criteria
Admin	can manage services , admins ,clients and clerks	Everything operates smoothly	Must log in using username and password fields
	View the appointment list	I can verify scheduled appointments	- Appointment list must be accessible via a designated box
	View the complaint list	I can ensure problems are being solved	Clerks must log complaints into the complaint box

As A/An	I want to	So that	Acceptance Criteria
Worker	give services		Must log in using username and password fields

	can see the appointments which are booked for the worker only		have to visit the worker dashboard

As A/An	I want to	So that	Acceptance Criteria
Clerk	Log in	I can start my work	I can successfully log in with my username and password
	Handle support tickets	I can help users with their problems	I can view and respond to tickets from clients and workers
	Manage booking statuses	I can track jobs in progress	I can change the status of a booking
	Help verify user profiles	New users can be approved	I can review profiles and mark them as ready for admin approval
	communicate with users	I can help them with routine questions	I can send and receive messages with clients and workers
	Update data	All information is correct	I can edit and save changes to user and service data
	Create reports	I can help the admin make decisions	I can generate basic reports from the system's data
	Organize user feedback	We can improve our services	I can see and sort feedback submitted by users

Trello Board

Existing Software Solutions

There are already some apps and websites that offer home service bookings, such as:

- **UrbanClap** – An India-based app for booking many kinds of home services.
- **TaskRabbit** – Helps people hire freelancers for local tasks and services.
- **Thumbtack** – A U.S. platform that connects customers with service professionals.
- **Handy** – Focuses on cleaning, repairs, and installations.
- **Housejoy** – An Indian platform for home cleaning, beauty, and repair services.

Software Development Process Model Selection

The model is Agile XP . Because it is -

- **Customer Focus** – Customers stay involved, so the platform always fits their needs.
- **Quick Updates** – Features like booking, payments, and verification can be built and improved fast.
- **Flexible** – Easy to adapt when new services or changes are needed.
- **High Quality** – Practices like testing and teamwork keep the system safe and reliable.
- **Simple & Fast** – Only build what's needed, avoiding extra complexity.

Project Environment Analysis

1. **Nature of Requirements** : Core needs (registration, booking, payments, worker matching) are stable. Extra features (helpline, complaint system) may change.

2. **Uncertainty** : High – user demands and market needs may change. XP handles this well with continuous feedback.
3. **Team Size** : Small to medium (4–8 members). XP suits small, collaborative teams.
4. **Communication Style** : Frequent face-to-face or virtual meetings, using tools like Trello .
- 5.

How XP Supports the Project

- **Pair Programming**: Two developers work together, which helps catch mistakes and improve code quality.
- **Test-Driven Development (TDD)**: Features like worker matching are tested before coding to reduce errors.
- **Continuous Integration (CI)**: New code is added every day to avoid big problems later.
- **Customer/Stakeholder Feedback**: Admins, staff, and clients give regular feedback to keep the platform aligned with their needs.
- **Sustainable Pace**: Work is done in small 1–2 week cycles, making progress steady and predictable.

Flexibility of Agile XP

- Can **add, remove, or update features** in each iteration.
Example: Adding new services like pest control later.
- Can **integrate new technologies** or tools easily.
Example: Adding a new payment gateway or real-time map service.
- Can **adapt to user feedback quickly**.
Example: Adding in-app chat or improved dashboards for clerks.
- Keeps the platform safe and working well while making changes.
- Helps meet the needs of **clients, workers, and admins** efficiently.

Solution

People struggle to quickly find trusted workers for home services. Our platform solves this by **matching clients with verified local service providers** using location-based search, secure payments, and admin-clerk oversight. This ensures **fast service, safety, and trust** for both clients and workers.

Target Users & Benefits

1. **Clients:** Quick, safe access to nearby verified workers.
2. **Workers:** Steady jobs, trusted profiles, secure payments.
3. **Admins/Clerks:** Easy monitoring, approvals, and dispute handling.

Scientific Contribution****

- **Algorithms:** Real-time location-based worker matching.
- **Verification system:** Structured worker trust model.
- **Data insights:** Service demand, performance, satisfaction analytics.

Model Selection (Extreme Programming – XP)

- **Rapid iterations:** Small releases ensure working software from early stages.
- **User involvement:** Continuous client/worker feedback guides features.
- **Pair programming & testing:** Improves quality and reduces technical risk

Risk Management

- **Frequent releases:** Reduce delivery risk.
- **Continuous testing:** Prevents defects early.
- **User feedback:** Handles changing requirements and market risks.

Relation to Project Schedule

Iteration 1: Login & profiles.

Iteration 2: Service request → worker matching.

Iteration 3: Booking system.

Iteration 4: Admin/clerk dashboards.

Iteration 5: Secure payment & feedback.

Iteration 6: Refinement & deployment

Justification for Model Choice.

Extreme Programming (XP) is ideal for this project because it delivers working software quickly, involves users continuously, ensures high code quality through testing and pair programming, and adapts easily to changing requirements.

1.3 Project Role Identification and Responsibilities

Main roles involved in the project

Development Roles

- **Frontend Developer** → Builds client & worker apps, user interface.
- **Backend Developer** → Handles service matching, booking, payments, database.
- **Tester** → Checks for bugs and ensures quality.

Management Roles

- **Project Manager** → Plans, coordinates, and tracks progress.
- **Business Analyst** → Collects requirements and clarifies features.
- **Admin/Clerk** → Manages users, verifies workers, handles support.

1. Requirements Gathering

- The **Business Analyst** talks to clients, workers, and admins to understand what they need from the system.
- The **Project Manager** defines the project's scope, sets the timeline, and plans resources.
- The **Admin/Clerk** shares real-world input on how worker verification and support should work.

2. Design

- The **Frontend Developer** sketches out how the apps and screens will look and flow.
- The **Backend Developer** designs the database structure and overall system logic.
- The **Business Analyst and Project Manager** review the design to make sure it matches the requirements.

3. Implementation

- The **Frontend Developer** builds the actual app screens and user features for clients and workers.
- The **Backend Developer** codes the service-matching, booking, and payment system.
- The **Admin/Clerk** tries out early versions and gives feedback to make sure the workflow is practical.

4. Testing

- The **Tester** checks the system thoroughly, running tests to find bugs and usability issues.
- The **Developers** fix the problems that come up during testing.

- The **Project Manager** makes sure testing is complete and all features work as planned.

5. Deployment

- The **Backend Developer** sets up the server and database for live use.
- The **Frontend Developer** publishes the mobile apps.
- The **Project Manager** oversees the rollout to ensure everything goes smoothly.
- The **Admin/Clerk** begins daily operations—approving workers, handling support, and monitoring the system.

Responsible for decision making, quality assurance and resource management

Decision Making

- **Project Manager** → Decides project priorities, scope, and timeline.
 - **Business Analyst** → Guides decisions on which features are needed.
 - **Admin** → Handles daily choices like worker approvals and dispute resolution.
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Quality Assurance

- **Tester** → Finds bugs and checks system quality.
 - **Developers** → Fix issues and keep the system stable.
 - **Project Manager** → Ensures proper testing is done before launch.
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Resource Management

- **Project Manager** → Manages people, time, and budget.
- **Backend Developer/System Admin** → Looks after servers, databases, and technical resources.
- **Admin/Clerk** → Manages worker data, support tasks, and daily operations.

Responsibilities are distributed among the team members

- **Project Manager** : Guides the team, makes decisions, and keeps the project on track because they are good at planning and coordination.
- **Business Analyst** : Understands what users need and translates it into features, bridging the gap between users and developers.
- **Frontend Developer**: Builds the app screens and user interface, using their UI/UX skills to make it easy to use.
- **Backend Developer/System Admin**: Handles servers, databases, and system logic, ensuring the platform works reliably.
- **QA Tester**: Finds bugs and checks usability, making sure the system works smoothly before launch.
- **Admin/Clerk**: Manages worker approvals, support, and daily operations, keeping everything running safely and efficiently.

2 Functional Requirements

Major Functionalities

- **Client Side**
 - Register and log in to create an account
 - Request services
 - Get matched with available workers nearby.
 - Receive booking confirmations .

- Access a helpline for support.
- **Worker Side**
 - Register and set up a service profile.
 - Manage availability and service areas.
 - View and accept service requests from clients.
- **Admin Side**
 - Approve worker registrations after verification.
 - Manage users, services, and disputes.
 - Monitor worker activities and platform usage.
- **Clerk Side**
 - Verify worker profiles and documents.
 - Handle support tickets and client/worker communication.
 - Collect feedback and prepare reports.
 - Assist admins with daily operations.

core services, operations and features

Core Services

- Home repairs (plumbing, electrical work, appliance fixing, etc.)
- Cleaning and maintenance services
- General utility work (small household tasks)

Core Operations

- Clients can request and book services.
- Workers can accept jobs and manage their availability.
- Admins and clerks can verify workers, manage users, and solve issues.

Core Features

- **Registration & Login** for clients, workers, and admins.
- **Service Request & Matching** to connect clients with the right worker.
- **Booking & Notifications** for job confirmation.
- **Profile Management** for clients and workers.
- **Secure Payments** for safe transactions.
- **Support & Feedback System** for resolving problems and improving services.